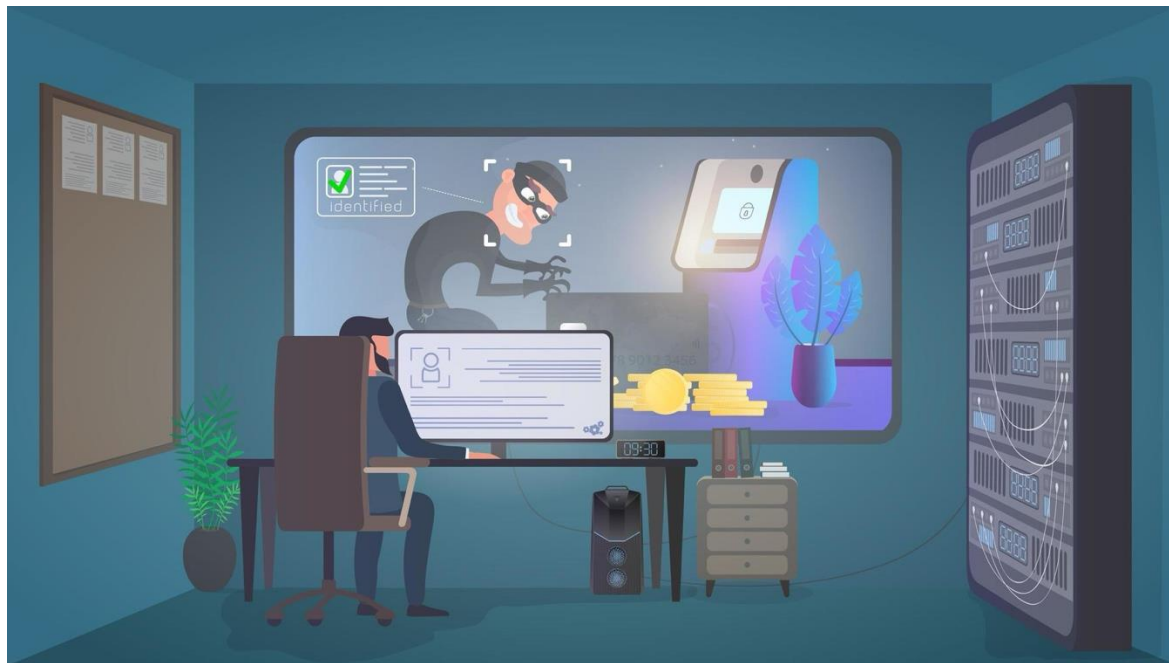




7. IDS

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IDS

- ◇ Intrusion Detection System
- ◇ To catch criminals in the cyberspace
- ◇ Means to identify intrusions



IDS

- ◆ Intrusion is...
 - ◆ A set of actions that compromises CIA
 - ◆ Actions include
 - ◆ Causing DoS
 - ◆ Installing Backdoors
 - ◆ Spreading Viruses
 - ◆ Exploiting software vulnerabilities
 - ◆ Etc.

IDS

- ◇ Intrusion detection is...
 - ◇ Detecting signs of security problems and analysing the events happening in a computer/network systems
 - ◇ We assume that system activities (e.g., users, program) can be monitored
 - ◇ Steps
 1. Monitor – analyse hosts/networks
 2. Identify – any misuse/abnormal activities
 3. Assess – severity and alert administrators

IDS

- ◇ Intrusion detection system is...
 - ◇ Combinations of software and hardware to perform intrusion detection
 - ◇ Can be placed in multiple locations in the network



IDS

- ◇ ID models
 - ◇ Misuse detection
 - ◇ Signature or knowledge-based
 - ◇ Anomaly detection
 - ◇ Behaviour or statistic-based
- ◇ Detect intrusion from
 - ◇ Hosts
 - ◇ Networks
 - ◇ Application
 - ◇ Hybrid



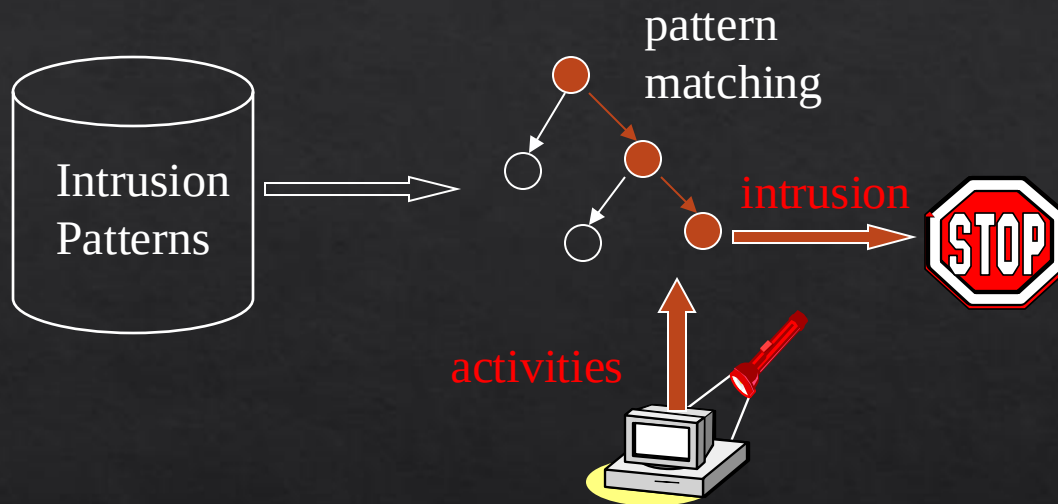
IDS

- ◇ Misuse detection
 - ◇ Well-known attack signatures are stored in the database
 - ◇ Current audit trails are compared with stored signatures
- ◇ Anomaly detection
 - ◇ Identify deviation from normal usage pattern
 - ◇ Apply statistical measures to classify activities/behaviours

Misuse Detection

- ◇ Find network, host traffic patterns that match known signatures
- ◇ Advantage:
 - ◇ Many attacks have distinct signatures
- ◇ Disadvantages:
 - ◇ IDS's signature database must be updated to keep pace with new attacks
 - ◇ Malicious code authors intentionally use tricks to fool these IDSs

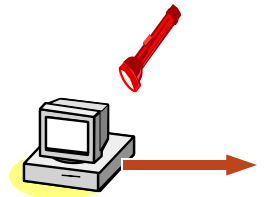
Misuse Detection



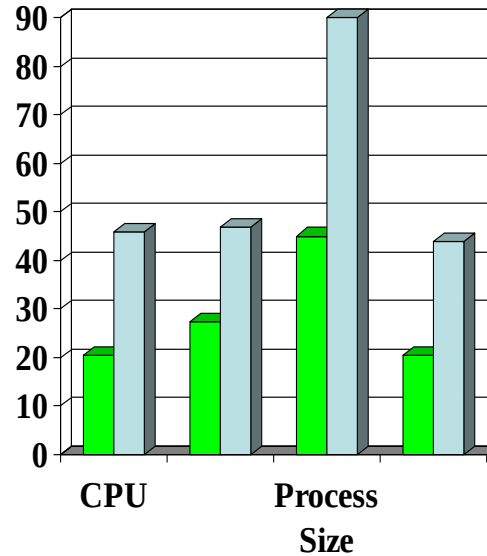
Anomaly Detection

- ◇ Statistical anomaly-based IDS sample network activity, compare to “known normal” traffic
- ◇ IDS sounds alarm when activity is outside baseline parameters
- ◇ Advantage:
 - ◇ IDS can detect new types of attacks
- ◇ Disadvantages:
 - ◇ Requires more overhead, compute power than signature-based IDSs
 - ◇ May generate many false positives

Anomaly Detection



activity
measures



probable
intrusion



IDS Terminology

- ◇ Alert, alarm : self-explanatory
- ◇ False negative : IDS fails to detect actual attack
- ◇ False positive : Attack alert when none occurred
- ◇ Confidence value : Estimate of attack probability
- ◇ Alarm filtering : self-explanatory

Key Performance Metrics

- ◇ Algorithm

- ◇ Alarm: A ; Intrusion: I

- ◇ Detection (true alarm) rate: $P(A | I)$

- ◇ False negative rate $P(\neg A | I)$

- ◇ False alarm rate: $P(A | \neg I)$

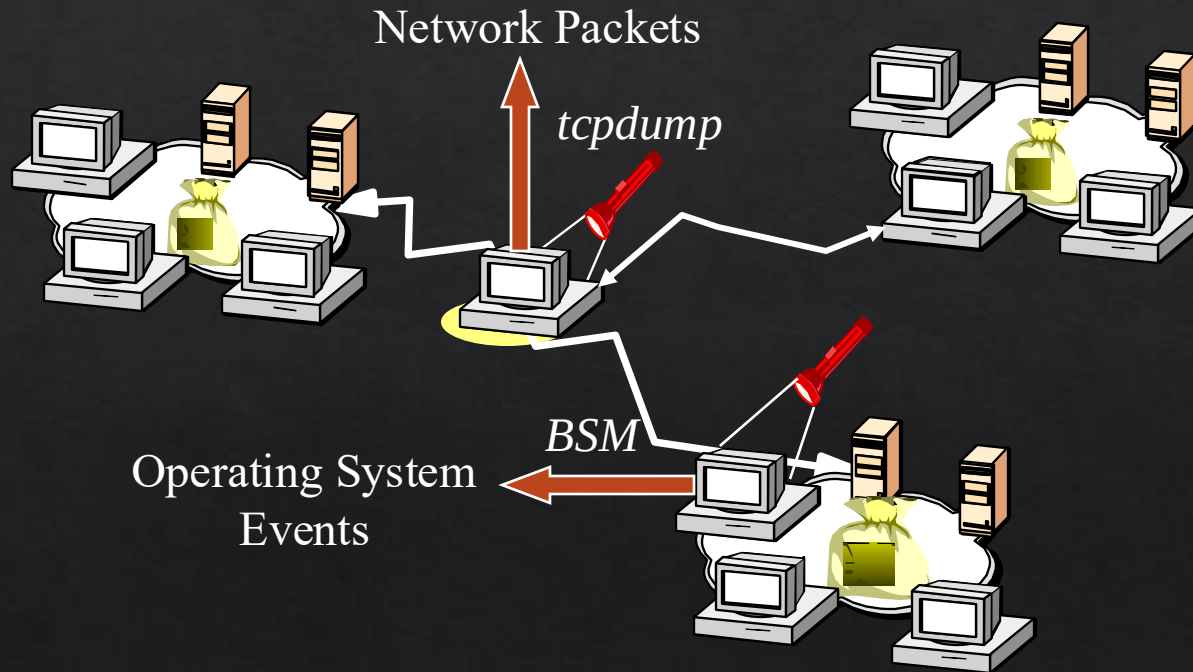
- ◇ True negative rate $P(\neg A | \neg I)$

- ◇ Architecture

- ◇ Scalable

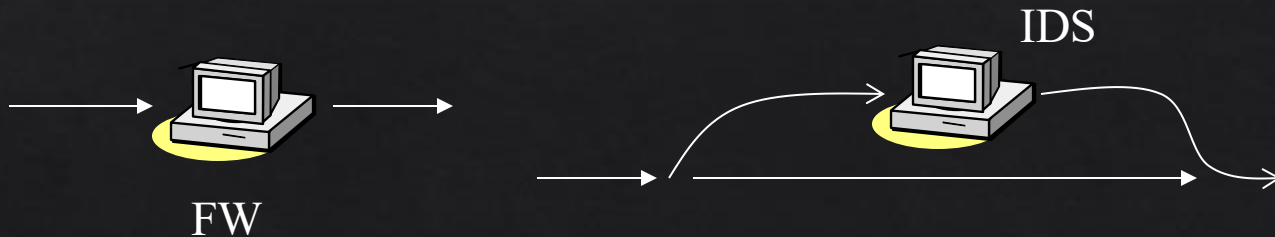
- ◇ Resilient to attacks

Monitoring Networks and Hosts



Firewall Versus Network IDS

- ◇ Firewall
 - ◇ Active filtering
 - ◇ Fail-close
- ◇ Network IDS
 - ◇ Passive monitoring
 - ◇ Fail-open



IDS Placement

- ◇ IDS placements – Host and Network
- ◇ HIDS (Host-based IDS)
 - ◇ Sits inside the host system
 - ◇ Monitor the system usage
 - ◇ OS dependent
 - ◇ Application IDS also belongs here, but focuses on applications only
- ◇ NIDS (Network-based IDS)
 - ◇ Sits in the network attached at a location
 - ◇ Capture and monitor packets in the network

NIDS

- ◊ Resides on computer or appliance connected to segment of an organization's network; looks for signs of attacks
- ◊ When examining packets, a NIDS looks for attack patterns
- ◊ Installed at specific place in the network where it can watch traffic going into and out of particular network segment

NIDS Signature Matching

- ◇ NIDSs look for attack patterns for detection
- ◇ Accomplished via certain implementation of TCP/IP stack:
 - ◇ Protocol stack verification: look for invalid packets
 - ◇ App. Protocol verification: look at higher-order protocols for unexpected behaviour or improper use

NIDS Advantages & Disadvantages

Advantages

- Org. can monitor large network with few devices
- Passive; deployment minimally disrupts operations
- Less susceptible to attack; attackers may not detect them

Disadvantages

- Can be overwhelmed by volume of network traffic
- Need to monitor *all* traffic
- Usually, cannot analyse encrypted network packets
- Cannot determine if attack was successful
- Cannot detect some attacks (e.g., fragmented packets)

HIDS

- ◇ HIDS runs on a particular computer, monitors activity only on that system
- ◇ Benchmarks, monitors key system files; detects when intruders' file I/O
- ◇ HIDSs work on principle of configuration management
- ◇ Unlike NIDSs, HIDSs can be installed to access info. that's encrypted in transit over network

HIDS Advantages & Disadvantages

Advantages

- Detect local events, attacks on host systems that NIDSs may not
- Can view encrypted traffic (as it has been decrypted on system)
- HIDSs unaffected by switched network protocols
- Can detect inconsistencies in apps, programs by examining audit logs

Disadvantages

- Harder to manage than NIDSs
- Vulnerable to attacks against host operating system, HIDS
- Cannot detect scans of multiple hosts, non-network devices
- HIDSs potential targets for denial-of-service (DoS) attack
- May use lots of disk space
- Possible large compute performance overhead on host systems

AppIDS

- ◇ Application-based IDS (AppIDS) looks at apps for abnormal events
- ◇ AppIDS may be configured to intercept requests:
 - ◇ File System
 - ◇ Network
 - ◇ Configuration
 - ◇ Process' Virtual Memory Address Space

AppIDS Advantages & Disadvantages

Advantages

- Aware of specific users; can observe interaction between apps and users
- Functions with encrypted incoming data

Disadvantages

- More susceptible to attack
- Less capable of detecting software tampering
- May be fooled by forms of spoofing

Current IDS direction

- ◇ The rise of machine learning (ML) has led IDS research and development based on ML techniques.
- ◇ Due to their effectiveness in terms of identifying anomalies in data, ML approaches became widely adopted technique for all IDS applications (network, host, application).
- ◇ However, practicality must be considered to ensure the IDS performance meets security requirements.

IDS in practice

- ◇ It is important to understand the suitability of different IDSes in practice, due to varying factors such as:
 - ◇ Environment
 - ◇ Threat
 - ◇ Asset
 - ◇ Users
- ◇ To better understand the practicality of IDSes, they need to be evaluated.
- ◇ Q: How to evaluate IDSes?
 - ◇ i.e., how could you select the most appropriate IDS for your needs?

IDS in practice

- ◇ Users must be careful when choosing the right IDS, as they often have issues, such as:
 - ◇ Code errors
 - ◇ Dependency errors
 - ◇ Unavailability of tool/code
 - ◇ High resource requirements
 - ◇ Etc.

NIDS	Year	Dataset	Source
Deep Neural Network (DNN) [18]	2018	KDDCup-'99'	Conference: ICCCNT
Kitsune [7]	2018	Custom IoT Dataset	Conference: NDSS
HELAD [17]	2020	CICIDS2017	Journal: MDPI Informatics
Multiclass Classification [20]	2020	ASNM Datasets	Conference: DSAA
ARTEMIS. [21]	2021	Custom Dataset	Conference: LATINCOM
Dense-Attention-LSTM, DAL [22]	2021	UNSW-NB15	Conference: IWCMC
I-SiamIDS [23]	2021	CICIDS, NSL-KDD	Journal: Applied Intelligence
SecureTea [24]	2021	N/A	GitHub
AutoML [25]	2022	CICIDS2017, IoTID20	Journal: Engineering Applications of Artificial Intelligence
Deep Belief Networks NIDS [26]	2022	CICIDS2017	Conference: SciSec
RIDS [27]	2022	Custom Dataset	Conference: GLOBECOM
StratosphereIPS (Slip) [19]	2022	N/A	GitHub
IDS-ML [28]	2022	CICIDS2017	Journal: Software Impacts
xNIDS [29]	2023	Mirai, CICDoS2017, NSL-KDD	Conference: USENIX Security
Suricata [30]	2023	N/A	GitHub

IDS in practice

- ◇ Even when they work, you might not get the results you expected because:
 - ◇ Optimised for different use cases
 - ◇ Performs poorly on out-of-domain applications
 - ◇ Input incompatibility
 - ◇ Evolving threats
 - ◇ Etc.

Dataset	Acc.	Prec.	Rec.	F1
IDS: Kitsune				
UNSW-NB15	0.6954	0.0221	0.2136	0.0401
BoT_IoT	0.9923	0.8153	0.8609	0.8375
CICIDS2017	0.5540	0.0109	0.9753	0.0216
Stratosphere	0.9921	0.9981	0.9027	0.9480
Mirai	0.8902	0.9999	0.8788	0.9354
Average:	0.8248	0.5693	0.7663	0.5565
IDS: HELAD				
UNSW-NB15	0.9717	0.0201	0.0107	0.0140
BoT_IoT	0.9793	0.6916	0.9011	0.7826
CICIDS2017	0.6437	0.9682	0.3706	0.5360
Stratosphere	0.9846	0.9805	1.0000	0.9902
Mirai	0.8898	0.9939	0.8786	0.9327
Average:	0.8938	0.7284	0.6322	0.6511
IDS: DNN				
UNSW-NB15	0.9820	0.9820	1.0000	0.9910
BoT_IoT	0.9770	0.9770	1.0000	0.9884
CICIDS2017	0.9800	0.9800	1.0000	0.9899
Stratosphere	0.2110	0.2110	1.0000	0.3485
Mirai	0.9060	0.9060	1.0000	0.9507
Average:	0.8112	0.8112	1.0000	0.8537
IDS: Slip				
UNSW-NB15	0.8735	0.0000	0.0000	0.0000
BoT_IoT	0.0018	0.0000	0.0000	0.0000
CICIDS2017	0.9370	0.0037	0.0447	0.0068
Stratosphere	0.6745	0.8809	0.4739	0.6163
Mirai	0.8040	0.1243	0.0159	0.0282
Average:	0.6582	0.2018	0.1069	0.1303

Conclusion

- ◆ Intrusion detection is one of the first key steps in identifying and mitigating threats.
- ◆ Several approaches are defined, all with benefits and limitations.
- ◆ Due to a large number of IDS tools available nowadays, it is important to consider the practical application that meets your requirements.
 - ◆ However, this turns out to be quite challenging, due to the lack of standardised evaluation of IDS solutions.
- ◆ Further research must be carried out, and you may need to customise IDS solutions to meet your security requirements.

References

- ◇ Some figures from UC Berkley
- ◇ IDS comparisons
 - ◇ <https://doi.org/10.48550/arXiv.2403.17458>